

LC³Net: Ladder context correlation complementary network for salient object detection

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ABSTRACT

Currently, the existing salient object detection methods based on convolutional neural networks commonly resort to constructing discriminative networks to aggregate high level and low level features. However, the potential contextual information is always difficult to be fully and reasonably utilized, which usually causes either the absence of useful features or the contamination of redundant features. To address these issues, we propose a novel ladder context correlation complementary network (LC³Net) in this paper, which is equipped with three crucial components. At the beginning, we propose a filterable convolution block (FCB) to assist the automatic collection of information on the diversity of initial features by skillfully inserting the atrous convolutions into independent branches. Besides, we propose a dense cross module (DCM) to facilitate the intimate aggregation of different levels of features by validly integrating the semantic information and detailed information of both adjacent and non-adjacent layers. Furthermore, we propose a bidirectional compression decoder (BCD) to help the progressive shrinkage of multi-scale features from coarse to fine by elaborately leveraging multiple pairs of alternating top-down and bottom-up feature interaction flows. Extensive experiments demonstrate the superiority of our method against 20 state-of-the-art methods.

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1. Introduction

Salient object detection, also known as saliency detection, is derived with the goal of detecting the most conspicuous and distinctive regions in the scene. Due to the capability and scalability, this technique has been widely applied in the tasks of computer vision, such as image parsing [1], image captioning [2], image smoothing [3], image segmentation [4,5], image retrieval [6,7], visual tracking [8–10], search engines [11] and many more [12,13].

Recently, benefiting from the explosive development of deep convolutional neural networks (CNNs), a lot of profound saliency detection methods based on CNNs are constantly emerging, and encouraging signs of progress have been made [14–31]. These methods of this kind strictly conform to the end-to-end encoder-decoder pattern, that is, the hierarchical features of different levels are first fixedly extracted from the encoder, and then appropriately propagated to the decoder for iterative optimization to predict the saliency map. At present, many existing methods commonly resort to strengthening the feature representation via

investigating components of the model, thereby constructing discriminative networks to aggregate high level features and low level features.

As is known to all, high level features carry rich semantic information that possesses the relatively clear background, while low level features preserve enough detailed information that possesses the relatively clear boundary. Because of the differences of receptive fields at different layers, there is a big gap between high level features and low level features. As a result, it remains dilemmas in the utilization of potential contextual information. Actually, whether a region is salient or not is decided heavily by the context of the target. In other words, the key of saliency detection lies in how to grasp the complementarity of contextual information harmoniously. However, the correlation of context is quite difficult to be fully and reasonably exploited, which will inevitably aggravate the misjudgment of detection. On the one hand, when the contextual information could not be fully utilized, it will usually lead to the lack of useful features. In this case, the detection of objects fails in tangibly identifying the contours. On the other hand, when the contextual information could not be reasonably utilized, it will usually bring about the pollution of redundant features. In this case, the detection of objects is prone to suffer from the interference of noises.

To tackle the aforementioned problems in the previously existing methods, we propose a novel network called ladder context

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correlation complementary network (LC³Net) in this paper, which explicitly explores the reliable inherent complementarity of correlated contextual information. Specifically, LC³Net is equipped with three crucial components, which are filterable convolution block (FCB), dense cross module (DCM) and bidirectional compression decoder (BCD). As for FCB, it is plug and play, and is located immediately and directly behind the backbone network. Except that, it also enjoys the merit of being simple yet practical. It has independent branches with the atrous convolutions. This block is placed to assist the automatic collection of information on the diversity of initial features. As for DCM, it has two sub-components according to the two sampling types, one is that with upsampling (DCM-U), and the other is that with downsampling (DCM-D). These two own the opposite and symmetrical structure, which fuses the features of each level with all other higher or lower levels with the semantic information and detailed information of both adjacent and non-adjacent layers. The two modes of this module are combined to facilitate the intimate aggregation of different levels of features. As for BCD, it has three distinguished sub-components, namely, BCD₁, BCD₂ and BCD₃. Each of them has a pair of alternating feature interaction flows on the pathway from top to down and the pathway from bottom to top in turn. The three parts of this decoder are merged to complete the interaction of features to help the progressive shrinkage of multi-scale features from coarse to fine. Accordingly, the local saliency cues can be captured by atrous convolutions with a small rate and the global saliency cues can be captured by atrous convolutions with a large rate in FCB, and contextual association and interaction can be centralized in DCM and BCD. In a word, the block, module and decoder play a role in succession from the perspective of feature collection, aggregation and shrinkage. Without bells and whistles, we report extensive experimental results on five datasets in terms of four evaluation metrics to demonstrate the superiority of our method against 20 state-of-the-art methods. As shown in Fig. 1, the saliency maps obtained by our method can better suppress the surrounding background noises and display the expected boundary contours in comparison with those obtained by other methods.

The main contributions of this paper are summarized as follows:

- A filterable convolution block (FCB) is proposed to dynamically collect the diversity information of initial features by skillfully inserting the atrous convolutions into independent branches.
- A dense cross module (DCM) is proposed to tightly aggregate different levels of features by validly integrating the semantic information and detailed information of both adjacent and non-adjacent layers.
- A bidirectional compression decoder (BCD) is proposed to gradually shrink multi-scale features until refinement by elaborately leveraging multiple pairs of alternating top-down and bottom-up feature interaction flows.
- Experiments on five datasets in terms of four evaluation metrics prove that our proposed ladder context correlation complementary network (LC³Net) achieves remarkable results compared with 20 state-of-the-art methods.

The remainder of this paper is organized as follows. In Section 2, a brief review of the related work is stated. Section 3 describes the proposed ladder context correlation complementary network in detail. Section 4 analyzes the experimental results to show the effectiveness and efficiency of the proposed method. The conclusion is given in Section 5.

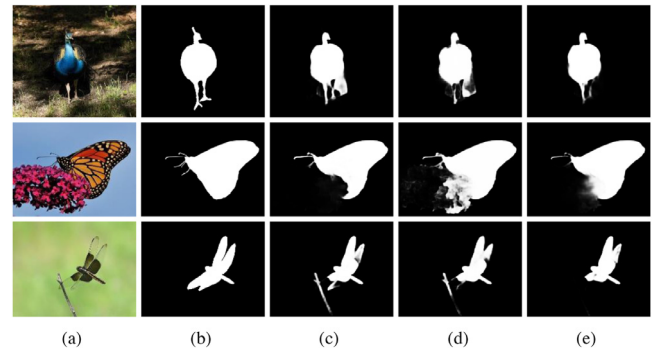


Fig. 1. Some saliency maps obtained by different methods. (a) Image; (b) Ground truth; (c) F³Net; (d) MINet; (e) Ours.

2. Related work

Early methods to detect salient objects depend purely on the hand-craft features and heuristic priors, receiving only limited performance. In recent years, thanks to the rise of deep learning, dramatic performance has been received in salient object detection. Here, we just concentrate on the saliency models on the basis of deep learning. For more comprehensive reviews, interested readers can refer to [32].

Throughout the past decades, hundreds of valuable saliency detection methods based on deep convolutional neural networks have been proposed. These methods are devoted to aggregating high level and low level features from the major aspects of short connection, attention mechanism, residual learning, gated unit and edge-aware knowledge. For instance, Zhang et al. [33] proposed to bridge all levels of features into the transport layer. Zhang et al. [34] proposed to degrade the checkerboard artifacts of deconvolution operators. Hou et al. [35] proposed to introduce the short connection into the skip-layer and predict the salient objects from saliency maps on each side-output. Luo et al. [36] proposed to capture both local and global cues through a multi-resolution grid structure. Deng et al. [37] proposed to alternately incorporate the features of shallow and deep layers to refine the predicted saliency maps. Chen et al. [38] proposed to assemble reverse attention to serve the side-output residual learning. Li et al. [39] proposed to borrow contour knowledge for saliency prediction. Zhang et al. [40] proposed to transfer information from deeper layers to shallower layers by recurrent connection. Zhang et al. [41] proposed to spread the information of multi-level features by using a gated bi-directional message passing structure. Wang et al. [42] proposed to recurrently locate salient objects with local saliency cues. Liu et al. [43] proposed to classify those capsules constructed from the image features into two streams. Su et al. [44] proposed to revisit the feature selectivity at boundaries and the feature invariance at interiors simultaneously. Zhao et al. [45] proposed to guide the saliency detection with the complementary information between the edge information and the object information. Wu et al. [46] proposed to abandon the low level features in the cascaded framework. Liu et al. [47] proposed to tap the potentials of pooling and add edge information. Zhu et al. [48] proposed to aggregate the attentional dilated convolutional features. Qin et al. [49] proposed to mix receptive fields of different sizes in the two-level nested U-structure. Zhao et al. [50] proposed to adaptively control the amount of information flowing from each encoder to each decoder. Wei et al. [51] proposed to iteratively feedback the features of different levels. Pang et al. [52] proposed to collaboratively learn knowledge guidance through mutual learning and self-learning. Sun et al. [53] proposed to improve the object location and boundary prediction

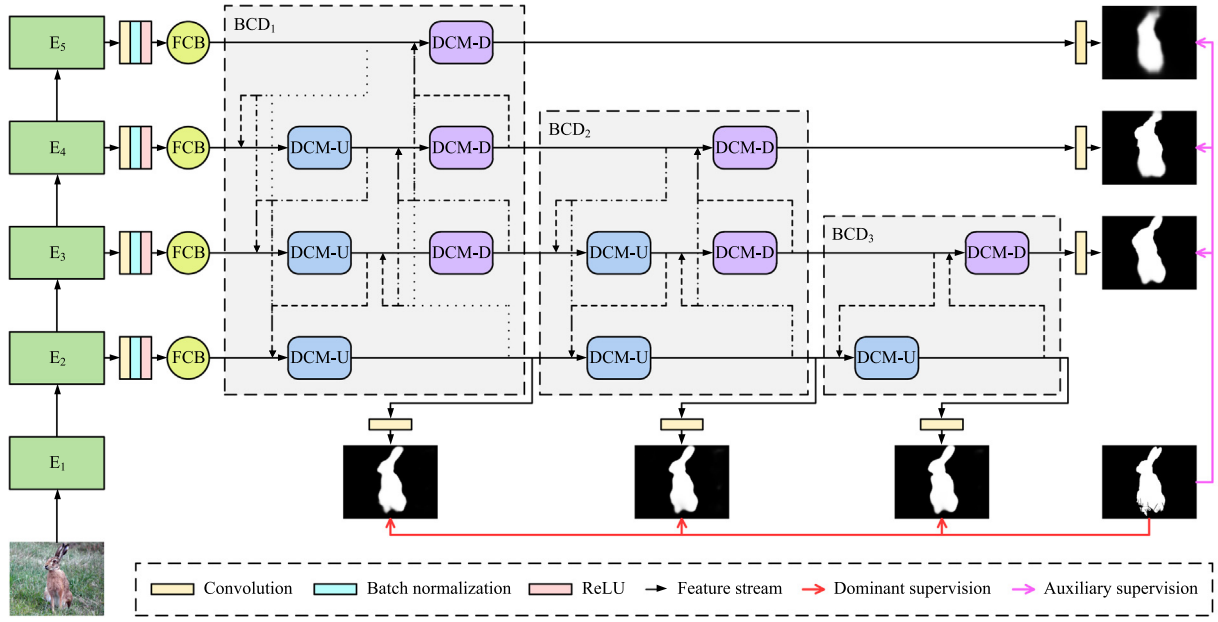


Fig. 2. The overall architecture of the proposed ladder context correlation complementary network (LC³Net). Several filterable convolution block (FCB) is placed to dynamically collect the diversity information of initial features. The two modes of several dense cross module (DCM), that with upsampling (DCM-U) and that with downsampling (DCM-D), are combined to tightly aggregate different levels of features. The three parts of several bidirectional compression decoder (BCD), BCD₁, BCD₂ and BCD₃, are merged to gradually shrink multi-scale features until refinement.

by additional downsampling and edge complement. Liu et al. [54] proposed to adopt new two-stream to narrow down the searching space when the routing from a low level capsule to the high level capsules.

Unlike the above methods, some methods are dedicated to probing the scope of context. Their consensus is that the global context enables the network to determine global saliency in the raw areas, and the local context enables the network to determine object local saliency in the meticulous areas. For instance, Zhao et al. [55] proposed to put the global context and local context of an image into a multi-context network. Liu et al. [56] proposed to selectively attend to the local or global pixel-wise contexts. Chen et al. [57] proposed to make full use of global context to capture the relationships between multiple salient objects or regions. Wang et al. [58] proposed to integrate the salient contextual features from sub-regions. Hu et al. [59] proposed to translate the spatial context over the entire feature map with different attenuation factors.

All these methods have exhibited sustainable and promising prospects. Nevertheless, there are still bottlenecks in dealing with the complex detection situations, which have plenty of space in utilizing contextual information. Motivated by these approaches, we intend to explicitly take advantage of the complementarity of correlated information between high level features and low level features.

3. Methodology

3.1. Network overview

Fig. 2 shows the overall architecture of the proposed ladder context correlation complementary network (LC³Net). For any backbone network, the feature f_i ($i = 1, 2, 3, 4, 5$) from different layers can be easily extracted by the encoder $\{E_1, E_2, E_3, E_4, E_5\}$. Owing to the fact that the feature f_1 imposes considerable computation cost, but the performance improvement is negligible, we only focus on the features f_2, f_3, f_4 and f_5 with the channels of {256, 512, 1024, 2048}. Through the combination of convolution,

batch normalization and ReLU operations, these features are uniformly squeezed to 64 channels and separately transmitted to the block FCB. Next, the features are successively propagated to the special decoders BCD₁, BCD₂ and BCD₃ neatly stacked with the multiple modules DCM-U and DCM-D. Afterwards, the 3×3 convolution is conducted to change the channel number to 1 for calculating the saliency scores.

3.2. Filterable convolution block

We propose the filterable convolution block (FCB) to dynamically collect the diversity information of initial features produced through the encoder. When it comes to portability, it is plug and play, attached to the back of the backbone network.

FCB is inspired by RFB [60] and modified from it. In contrast, there are five independent individual branches and each one has two steps organized in FCB. Although the idea is of simplicity, it has pretty good practicability.

For the convenience of narration, only the appearance of the core system of FCB is depicted here, as shown in Fig. 3. On the vertical line, each branch is assigned the atrous convolutions with different dilation rates or kernel sizes, or both. These branches are connected in parallel, and do not affect each other. They share the same input feature and each of them becomes a member of the output feature. On the horizontal line, the two steps are carried out in sequence. In the first step, all branches possess a convolution operation with the same dilation rate of 1, but the kernel sizes of those are 1×1 , 3×3 , 5×5 , 7×7 and 9×9 , respectively. In the second step, all branches possess a convolution operation with the same kernel size of 3×3 , but the dilation rates of those are 1, 3, 5, 7 and 9, respectively.

At the end of the core process, the collected information will be further concatenated and sent to a combination of convolution, batch normalization and ReLU operations.

3.3. Dense cross module

We propose the dense cross module (DCM) to tightly aggregate different levels of features, which powerfully promotes the

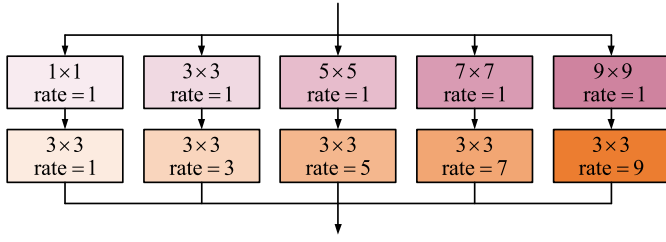


Fig. 3. Core system of the proposed filterable convolution block (FCB).

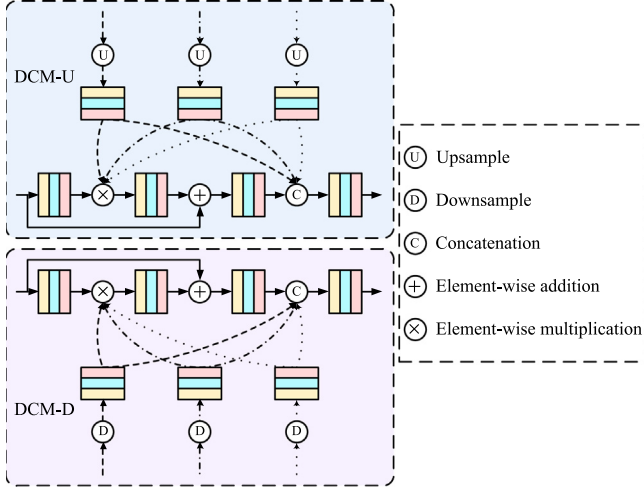


Fig. 4. Illustrate of two modes of the proposed dense cross module (DCM).

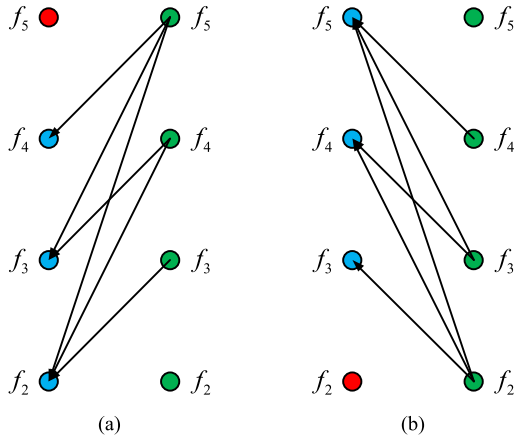


Fig. 5. The feature association rules in upsampling and downsampling phases of DCM-U and DCM-D. (a) Upsampling phase; (b) Downsampling phase.

integration of the semantic information and detail information of both adjacent and non-adjacent layers, so as to maximize the mining of useful features.

DCM consists of two modes according to the two sampling types, that with upsampling (DCM-U) and that with downsampling (DCM-D), as shown in Fig. 4. In particular, the former is committed to delivering high level features containing semantics to low level ones as much as possible, while the latter is committed to returning low level features containing details to high level ones as much as possible.

From the form of structure, DCM-U and DCM-D are just opposite and symmetrical. In both of them, the feature at each level is fused with all the features at higher or lower levels that

have to be upsampled or downsampled. Such fusions go through an element-wise multiplication, an element-wise addition and a concatenation operation, as well as a series of combinations of convolution, batch normalization and ReLU operations. The feature association in the upsampling and downsampling phases complies with the established rules, which is shown in Fig. 5. Informally, in the upsampling phase, the fifth layer features are isolated, while in the downsampling phase, the second layer features are isolated, because they have no corresponding features higher or lower than them in the sampling stage they belong to.

The specific process of DCM-U and DCM-D can be formulated as:

$$\hat{f}_i = \bigcup_i^4 (\Gamma(\text{Up}(f_{i+1}))), i = 4, 3, 2, \quad (1)$$

$$f'_i = \Gamma(\text{Concat}(\Gamma(\Gamma(f_i) \otimes \hat{f}_i) \oplus f_i), \hat{f}_i)), i = 4, 3, 2, \quad (2)$$

and

$$\hat{f}_i = \bigcup_2^{i-1} (\Gamma(\text{Down}(f_i))), i = 3, 4, 5, \quad (3)$$

$$f'_i = \Gamma(\text{Concat}(\Gamma(\Gamma(f_i) \otimes \hat{f}_i) \oplus f_i), \hat{f}_i)), i = 3, 4, 5, \quad (4)$$

where f_i , f'_i and $\hat{f}_i$ ($i = 2, 3, 4, 5$) represent the input feature, output feature and feature set of certain features after upsampling or downsampling of i th layer, respectively, $\bigcup$ is the union set, Up , Down and Concat denote the upsample, downsample and concatenation operations, respectively, Γ is the combination of convolution, batch normalization and ReLU operations, $\otimes$ denotes element-wise multiplication and $\oplus$ denotes element-wise addition.

DCM differs from GCM [61] in the usage of features at each level. The most obvious difference is that DCM only requires one round of upsampling and downsampling interactions of up to three layers, while GCM requires more overhead, which occupies lots of time in splitting.

3.4. Bidirectional compression decoder

We propose the bidirectional compression decoder (BCD) to shrink multi-scale features from coarse to fine in a progressive way, which is essentially a triple decoder and is built upon DCM.

For the features that need to be concerned (i.e., f_2, f_3, f_4 and f_5), they are thinly divided into three groups, $G_1 = \{f_2, f_3, f_4, f_5\}$, $G_2 = \{f_2, f_3, f_4\}$ and $G_3 = \{f_2, f_3\}$, in which f_2 is treated as the basic point, f_3, f_4 and f_5 are selectively less and less associated with it. G_1, G_2 and G_3 retain abundant multi-scale features and are correspondingly involved in three important parts of BCD, namely, $\text{BCD}_1, \text{BCD}_2$ and BCD_3 , as shown in Fig. 6. In each part, DCM-U and DCM-D appear in pairs, and excavate feature relying on the top-down and bottom-up manner in turn.

Taking BCD_1 as an example, it assumes that f_i ($i = 2, 3, 4, 5$) is the original feature and f'_i ($i = 2, 3, 4, 5$) is the updated feature. The fourth layer feature is first updated after the original f_4 and f_5 pass through DCM-U for the first time, and the third layer feature is then updated after the original f_3 and f_5 along with the new f_4 pass through DCM-U for the second time, and the third second feature continues to be updated after the original f_2 and f_5 along with the new f_4 and f_3 pass through DCM-U for the third time. So far, half of the procedure with DCM-U as the carrier is accomplished on the pathway from top to down, the other half of the procedure replaced by DCM-D as the carrier on the pathway from bottom to top is exactly reversed. A similar set of complete procedures also act on BCD_2 and BCD_3 .

The interactive relationship between $\text{BCD}_1, \text{BCD}_2$ and BCD_3 is shown in Fig. 7, which generally looks like a three-tier ladder. It is

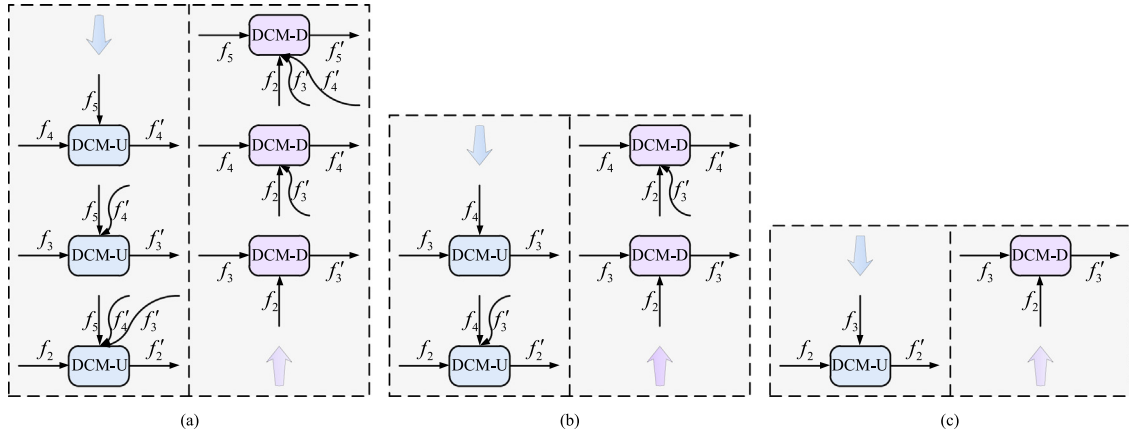


Fig. 6. Explanation of three parts of the proposed bidirectional compression decoder (BCD). (a) The schematic diagram of BCD_1 ; (b) The schematic diagram of BCD_2 ; (c) The schematic diagram of BCD_3 .

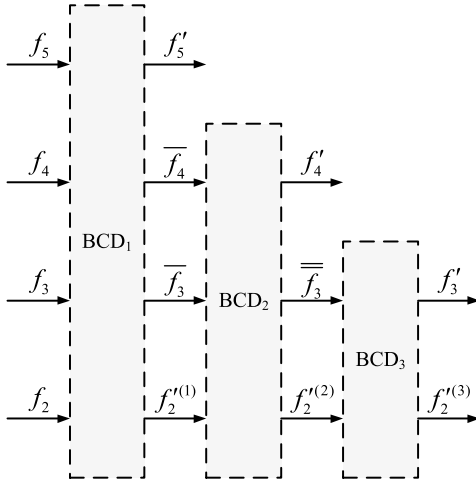


Fig. 7. The interactive relationship between BCD_1 , BCD_2 and BCD_3 .

worth pointing out that the design of three distinct parts instead of three identical parts can alleviate the interference of redundant features on the premise of ensuring the sufficient interaction of features. Meanwhile, the size of the model is able to be slightly compacted with fewer computational parameters in this way.

Let f_i ($i = 2, 3, 4, 5$), f'_i ($i = 2, j = 1, 2, 3$) and $\bar{f}_i$ ($i = 3, 4, 5$) be the input feature, the t th part of output feature, the tail of output feature of i th layer, respectively. The rough process of feature interaction in the three parts can be formulated as:

$$\begin{aligned} (f'_2(1), \bar{f}_3, \bar{f}_4, f'_5) &= BCD_1(f_5, f_4, f_3, f_2) \\ (f'_2(2), \bar{f}_3, f'_4) &= BCD_2(\bar{f}_4, \bar{f}_3, f'_2(1)), \\ (f'_2(3), f'_3) &= BCD_3(\bar{f}_3, f'_2(2)) \end{aligned} \quad (5)$$

where $\bar{f}_i$ ($i = 3, 4$) and $\bar{f}_i$ ($i = 3$) represent the first interaction intermediate feature and second interaction intermediate feature of i th layer, respectively.

Once the input features are acquired, after discarding all intermediate features, here three output features in the three parts of the second layer will be obtained, which are regarded as the dominant stream features, and three output features in the three tails of the fifth, fourth and third layers will be obtained, which are regarded as the auxiliary stream features.

3.5. Loss function

In our network, the binary cross entropy (BCE) loss and intersection over union (IoU) loss are jointly employed for collaborative supervision. BCE loss convergences for all pixels regardless of the labels, and IoU loss faces the degree of overlap of images. They enforce local constraint and global constraint in calculating losses, respectively.

BCE loss and IoU loss can be expressed as:

$$\mathcal{L}_{bce} = - \sum_{p \in P, g \in G} (g \log p + (1 - g) \log(1 - p)), \quad (6)$$

and

$$\mathcal{L}_{iou} = 1 - \frac{\sum_{p \in P, g \in G} (p \times g)}{\sum_{p \in P, g \in G} (p + g - p \times g)}, \quad (7)$$

where $P = \{p | 0 < p < 1\}$ and $G = \{g | 0 < g < 1\}$ denote the prediction and ground truth, respectively.

In practice, several predictions are generated over multiple parts of the same layer together with even single tail of different layers. By decomposing these predictions, the spatial dependence is coupled, which can be flexibly regulated. That is to say, there are two categories of losses derived from the dominant streams and auxiliary streams, corresponding to dominant losses and auxiliary losses. The dominant losses gather the loss from the second layer, and the auxiliary losses gather the loss from the third to the fifth layers.

Therefore, dominant losses and auxiliary losses can be computed as:

$$\mathcal{L}_{dom}^i = \frac{1}{3} \sum_{j=1}^3 (\mathcal{L}_{bce}(P_i^{(j)}, G) + \lambda \mathcal{L}_{iou}(P_i^{(j)}, G)), i = 2, \quad (8)$$

and

$$\mathcal{L}_{aux}^i = \mathcal{L}_{bce}(P_i, G) + \lambda \mathcal{L}_{iou}(P_i, G), i = 3, 4, 5, \quad (9)$$

where λ is the hyperparameter that balances the two penalty terms and is set to 1 for simplicity, $P_i^{(j)}$ and P_i denote the prediction from the j th part of the i th layer and the prediction from the tail of the i th layer, respectively.

Finally, the total loss function of the network can be written as:

$$\mathcal{L}_{total} = \sum_{i=2}^2 \eta_i \mathcal{L}_{dom}^i + \mu \sum_{i=3}^5 \eta_i \mathcal{L}_{aux}^i, \quad (10)$$

where μ is a trade-off parameter set to 1 empirically, η_i ($i = 2, 3, 4, 5$) are the weight coefficients, which are regularly set to 1, 1/2, 1/4 and 1/8, respectively.

Table 1
Quantitative comparison of different methods on five datasets in terms of four evaluation metrics.

Method	ECSSD				DUT-OMRON				PASCAL-S				HKU-IS				DUTS-TE			
	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$
Amulet [33]	0.893	0.915	0.912	0.059	0.780	0.743	0.784	0.098	0.820	0.837	0.829	0.098	0.884	0.897	0.914	0.051	0.803	0.778	0.798	0.085
UCF [34]	0.883	0.911	0.888	0.078	0.758	0.735	0.763	0.132	0.803	0.830	0.787	0.126	0.866	0.888	0.893	0.074	0.777	0.771	0.766	0.117
DSS [35]	0.882	0.916	0.924	0.053	0.788	0.772	0.846	0.066	0.797	0.836	0.851	0.096	0.880	0.900	0.927	0.050	0.824	0.825	0.885	0.056
NLDF [36]	0.875	0.905	0.912	0.063	0.770	0.753	0.817	0.080	0.804	0.833	0.842	0.099	0.878	0.902	0.929	0.048	0.816	0.813	0.855	0.065
R ³ Net [37]	0.910	0.934	0.940	0.040	0.817	0.795	0.859	0.062	0.805	0.843	0.845	0.094	0.895	0.915	0.945	0.035	–	–	–	–
RAS [38]	0.893	0.921	0.922	0.056	0.814	0.787	0.849	0.062	0.795	0.838	0.837	0.104	0.887	0.913	0.931	0.045	0.839	0.831	0.864	0.059
PAGR [40]	0.889	0.927	0.917	0.061	0.775	0.771	0.843	0.071	0.818	0.856	0.854	0.093	0.889	0.919	0.941	0.047	0.838	0.854	0.883	0.056
PiCANet [56]	0.917	0.935	0.927	0.046	0.832	0.803	0.848	0.065	0.854	0.870	0.862	0.076	0.905	0.919	0.941	0.044	0.869	0.860	0.873	0.051
BANet [44]	0.924	0.945	0.953	0.035	0.832	0.803	0.865	0.059	0.852	0.875	0.887	0.070	0.913	0.930	0.955	0.032	0.879	0.872	0.907	0.040
SCRN [62]	0.927	0.950	0.942	0.037	0.837	0.811	0.869	0.056	0.868	0.887	0.887	0.064	0.917	0.935	0.954	0.033	0.885	0.887	0.901	0.040
EGNet [45]	0.925	0.947	0.947	0.037	0.841	0.815	0.874	0.053	0.852	0.875	0.877	0.075	0.918	0.935	0.956	0.031	0.887	0.889	0.907	0.039
PAGE [63]	0.912	0.931	0.943	0.042	0.824	0.791	0.860	0.062	0.839	0.857	0.878	0.078	0.904	0.920	0.948	0.036	0.854	0.838	0.886	0.052
AFNet [64]	0.913	0.935	0.941	0.042	0.826	0.797	0.859	0.057	0.849	0.868	0.886	0.071	0.906	0.925	0.949	0.036	0.867	0.863	0.895	0.046
CPD [46]	0.918	0.939	0.949	0.037	0.825	0.797	0.873	0.056	0.847	0.869	0.887	0.072	0.906	0.925	0.952	0.034	0.869	0.865	0.904	0.043
PoolNet [47]	0.926	0.949	0.948	0.035	0.831	0.805	0.874	0.054	0.866	0.890	0.888	0.065	0.919	0.936	0.959	0.030	0.886	0.889	0.912	0.037
CapSal [65]	0.826	0.862	0.866	0.074	0.674	0.639	0.703	0.096	0.838	0.866	0.878	0.073	0.850	0.884	0.907	0.058	0.815	0.823	0.866	0.062
BASNet [66]	0.916	0.942	0.921	0.037	0.836	0.805	0.869	0.056	0.837	0.862	0.852	0.077	0.908	0.930	0.947	0.033	0.866	0.859	0.884	0.048
GateNet [50]	0.920	0.945	0.943	0.040	0.838	0.818	0.868	0.055	0.857	0.880	0.883	0.069	0.915	0.933	0.954	0.033	0.885	0.888	0.903	0.040
F ³ Net [51]	0.924	0.945	0.946	0.033	0.838	0.813	0.876	0.053	0.860	0.883	0.893	0.063	0.917	0.937	0.959	0.028	0.888	0.891	0.918	0.035
MINet [52]	0.925	0.947	0.953	0.033	0.833	0.810	0.873	0.055	0.856	0.879	0.898	0.064	0.920	0.935	0.961	0.028	0.884	0.884	0.917	0.037
LC ³ Net (ours)	0.925	0.951	0.953	0.032	0.837	0.819	0.878	0.051	0.866	0.891	0.904	0.059	0.921	0.942	0.963	0.028	0.893	0.898	0.928	0.033

↑ indicates that larger is better and ↓ indicates that smaller is better. The best results are marked in **bold**. “–” means the authors have not provided corresponding saliency maps.

4. Experiments

4.1. Experimental setup

Datasets. We choose five popular benchmark datasets, which are ECSSD [67], DUT-OMRON [68], PASCAL-S [69], HKU-IS [70] and DUTS [71]. These datasets are composed of 1000, 5168, 850, 4447 and 15572 images and their annotations, respectively. DUT-TR, the training set of DUTS, has 10553 images and corresponding annotated maps, while DUTS-TE, the testing set of DUTS, has 5019 images and corresponding annotated maps. Among them, DUT-TR is used for training, while the rests are used for testing. Note that only 1447 images and maps in HKU-IS are used following [35,38,70].

Evaluation Metrics. We select four typical evaluation metrics, which are S-measure (S_α) [72], maximum F-measure (F_β) [73], E-measure (E_ξ) [74] and mean absolute error ($\mathcal{M}$) [75]. They are defined as:

$$S_\alpha = \frac{S_o + S_r}{2}, \quad (11)$$

where S_o and S_r assess object-aware and region-aware structure similarities,

$$F_\beta = \frac{(1 + \beta^2) \times \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}}, \quad (12)$$

where β^2 is set as 0.3 to emphasize more on the precision than recall suggested in [32],

$$E_\xi = \frac{1}{W \times H} \sum_{x=1}^W \sum_{y=1}^H \phi(x, y), \quad (13)$$

where W and H are width and height, and ϕ is the enhanced alignment matrix,

$$\mathcal{M} = \frac{1}{W \times H} \sum_{x=1}^W \sum_{y=1}^H |P(x, y) - G(x, y)|, \quad (14)$$

where P and G denote the prediction and ground truth. Additionally, we also plot the precision–recall curve and F-measure curve.

Implementation Details. All experiments are implemented on an NVIDIA GeForce RTX 2080 Ti GPU with 11 GB of memory using the PyTorch repository. Data augmentation technology adopts horizontal flipping, random cropping and multi-scale input images. ResNet-50 [76], pre-trained on ImageNet [77], is employed as the backbone network. The model is optimized by stochastic gradient descent (SGD) with a batch size of 32, a momentum of 0.9 and a weight decay of $5e-4$. The warm-up and decay strategies are utilized to adjust the learning rate under the maximum learning rate of 0.05. The epoch is set to 50 by default. During inference, each image is resized to 352×352 . Any post-processing operations (e.g., CRF [78]) are not applied. It takes about 2 h to train the model, and the test time for each image is about 0.015 s. The model memory is totally saved as 181 MB.

4.2. Comparison with state-of-the-arts

We compare the proposed method with 20 state-of-the-art methods, including Amulet [33], UCF [34], DSS [35], NLDF [36], R³Net [37], RAS [38], PAGR [40], PiCANet [56], BANet [44], SCRNet [62], EGNet [45], PAGE [63], AFNet [64], CPD [46], PoolNet [47], CapSal [65], BASNet [66], GateNet [50], F³Net [51] and MINet [52]. For fair comparison, the saliency maps of competing methods are provided by the authors or generated by the authorized codes. If the competitors has multiple versions based on different backbone networks (e.g., VGG-16 [79], ResNet-50 [76] and ResNeXt-101 [80]), we give priority to using the same version as ours.

Quantitative Comparison. The results of quantitative comparison are listed in Table 1. As can be seen, the performance of our method is outstanding on all datasets, especially on ECSSD, HKU-IS and DUTS-TE, in terms of all evaluation metrics. Notably, our method reaches 0.059 on PASCAL-S with respect to $\mathcal{M}$, which surpasses other methods by a large margin. Apart from that, our method ranks first in almost all cases. In addition, the precision–recall and F-measure curves are shown in Fig. 8. It can be seen that the curves of our method are more stable and higher than those of others, which can also deduce the leading performance.

Qualitative Comparison. The results of qualitative comparison are shown in Fig. 9. A variety of challenging scenarios are picked, in which the salient objects vary from the contrast, texture, size,

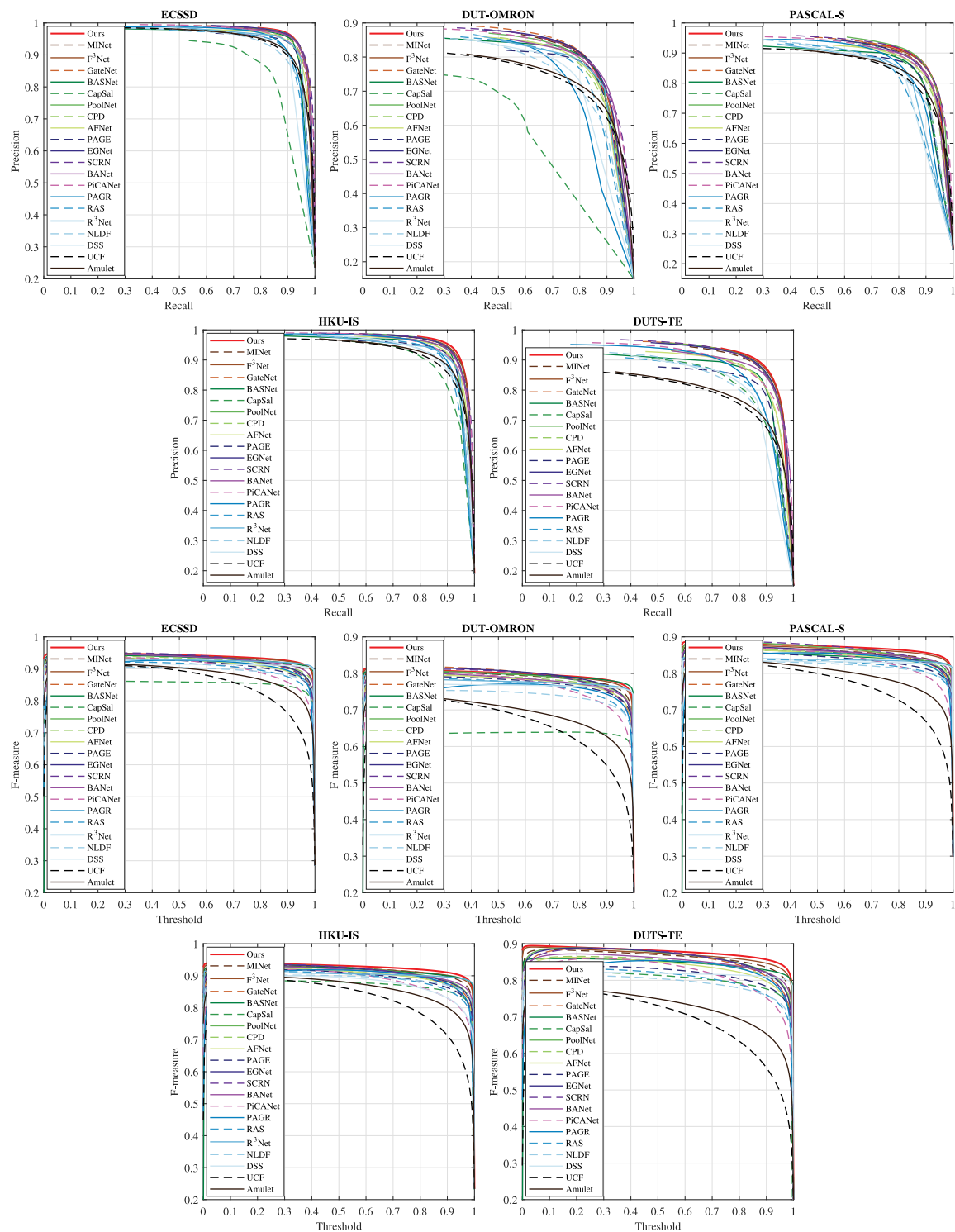


Fig. 8. The precision-recall and F-measure curves of different methods on five datasets. The first two rows show precision-recall curve and the last two rows show F-measure curve.

number and so on. Compared to other methods, our method can recognize coherent saliency objects with less false detection and attain more excellent performance. Apparently, our method is capable of handling these complex situations accurately, covering low contrast (i.e., 1st and 2nd rows), cluttered background (i.e., 3rd and 4th rows), small object (i.e., 5th and 6th rows) and multiple object (i.e., 7th and 8th rows). In short, these visualization results strongly verify the favorable robustness of our method to the various salient objects.

Comparison on SOC. Most recently, one of the most challenging datasets, SOC [82], was released. The dataset is composed of training set with 3600 pictures, validation set with 1200 pictures and test set with 1200 pictures. Following [81], we use 2400 images and maps from it to test on our trained model. At the same time, numerous methods have not yet yielded results on it, hence we only compare with the methods that publish results. The results of comparison on SOC are listed in Table 2. Obviously, our method outperforms all comparison methods in terms of all

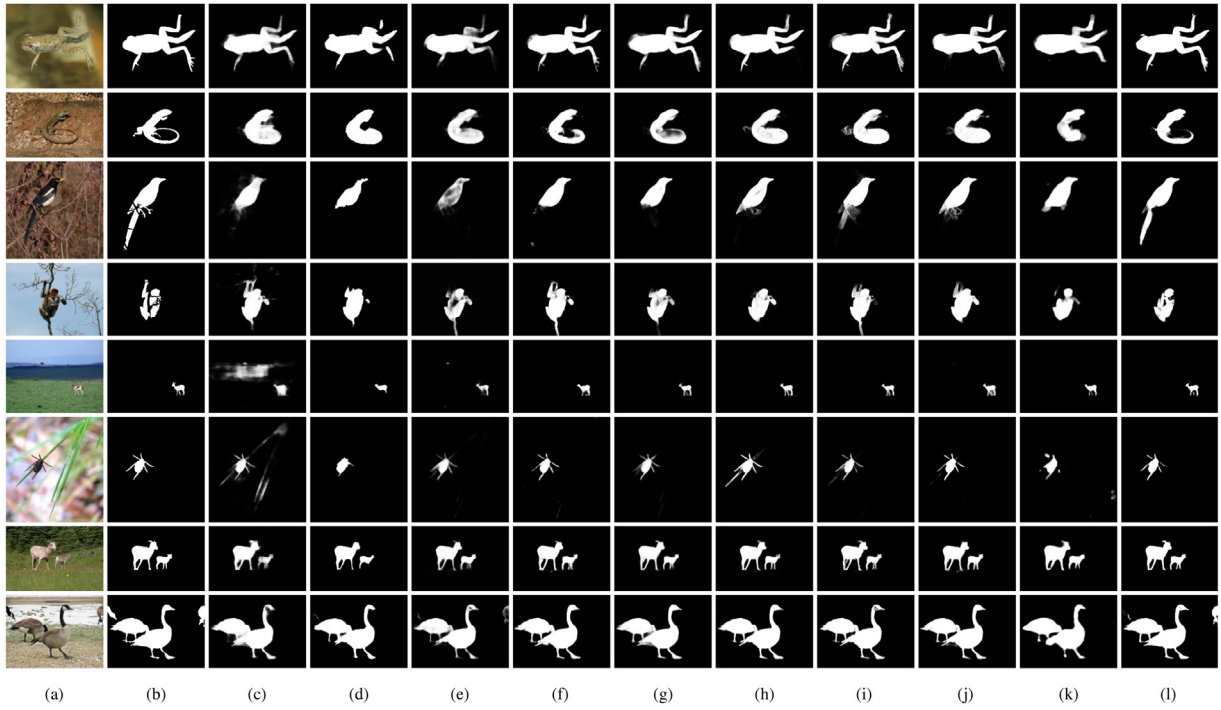


Fig. 9. Qualitative comparison of different methods in some representative scenes. (a) Image; (b) Ground truth; (c) Amulet; (d) DSS; (e) RAGR; (f) BANet; (g) SCRNet; (h) EGNet; (i) AFNet; (j) PoolNet; (k) CapSal; (l) Ours.

Table 2

Quantitative comparison of different methods on SOC in terms of four evaluation metrics.

Method	SOC			
	$S_{\alpha} \uparrow$	$F_{\beta} \uparrow$	$E_{\xi} \uparrow$	$\mathcal{M} \downarrow$
DSS [35]	0.673	0.628	0.759	0.141
NLDF [36]	0.723	0.666	0.784	0.129
Deepside [81]	0.781	0.739	0.833	0.093
LC ³ Net (ours)	0.796	0.808	0.895	0.079

Table 3

Impact of each component of the proposed method on DUT-OMRON and DUTS-TE in terms of four evaluation metrics.

	DUT-OMRON				DUTS-TE			
	$S_{\alpha} \uparrow$	$F_{\beta} \uparrow$	$E_{\xi} \uparrow$	$\mathcal{M} \downarrow$	$S_{\alpha} \uparrow$	$F_{\beta} \uparrow$	$E_{\xi} \uparrow$	$\mathcal{M} \downarrow$
Baseline	0.784	0.761	0.843	0.069	0.827	0.841	0.887	0.050
+B	0.821	0.776	0.850	0.064	0.874	0.864	0.903	0.046
+B+M	0.831	0.807	0.868	0.058	0.886	0.882	0.917	0.039
+B+M+D	0.837	0.819	0.878	0.051	0.893	0.898	0.928	0.033

"B" = FCB, "M" = DCM and "D" = BCD.

evaluation metrics, which implies that our method performs well even on the tricky dataset.

4.3. Ablation study

We study the impact of each component and sub-component in the proposed method.

Impact of Each Component. Here, we consider the schemes of "+B", "+B+M" and "+B+M+D", which stands for adding the individual or combinations of the proposed block, module and decoder into the baseline. The baseline refers to the network like feature pyramid networks (FPNs) [83]. The results of impact of each component are listed in Table 3. We can find that the

Table 4

Impact of DCM-U and DCM-D on DUT-OMRON and DUTS-TE in terms of four evaluation metrics.

	DUT-OMRON				DUTS-TE			
	$S_{\alpha} \uparrow$	$F_{\beta} \uparrow$	$E_{\xi} \uparrow$	$\mathcal{M} \downarrow$	$S_{\alpha} \uparrow$	$F_{\beta} \uparrow$	$E_{\xi} \uparrow$	$\mathcal{M} \downarrow$
w/o M	0.795	0.769	0.846	0.067	0.841	0.854	0.890	0.046
w/o M-U	0.803	0.779	0.849	0.065	0.843	0.868	0.899	0.043
w/o M-D	0.823	0.794	0.858	0.054	0.884	0.883	0.912	0.036
w/ M	0.837	0.819	0.878	0.051	0.893	0.898	0.928	0.033

"M-U" = DCM-U, "M-D" = DCM-D, and "w/ ★" and "w/o ★" mean with/without ★.

performance is continuously improved with the embedding of the increasing components including FCB, DCM and BCD, and the optimal performance is obtained by using all components, which reveals that each component is necessary to realize this method. Moreover, comparing our method with the baseline, we obtain gains of 5.3% for S_{α} , 5.8% for F_{β} , 3.5% for E_{ξ} and 1.8% for $\mathcal{M}$ on DUT-OMRON, and gains of 6.6% for S_{α} , 5.7% for F_{β} , 4.1% for E_{ξ} and 1.7% for $\mathcal{M}$ on DUTS-TE, respectively.

Impact of Each Sub-component. Here, we should try to consider two sets of schemes. The one thing, we consider the first set of schemes of "w/o M", "w/o M-U", "w/o M-D" and "w/ M", which examines our method of assembling the proposed module or its sub-component. The results of impact of DCM-U and DCM-D are listed in Table 4. We can observe that the "w/ M" scheme performs best, followed by the "w/o M-D" scheme and the "w/o M-U" scheme, and the "w/o M" scheme performs worst. Moreover, the "w/o M-U" scheme has a slight enhancement over the "w/o M" scheme, whereas the "w/o M-D" scheme has an evident enhancement over the "w/o M" scheme, indicating the overwhelming significance of DCM-U than DCM-D, despite both are vital. The other thing, we consider the second set of schemes of " $\Sigma = 1$ ", " $\Sigma = 2$ " and " $\Sigma = 3$ ", which examines our method of assembling the proposed decoder or its sub-component. The

Table 5

Impact of BCD₁, BCD₂ and BCD₃ on DUT-OMRON and DUTS-TE in terms of four evaluation metrics.

	DUT-OMRON				DUTS-TE			
	$S_{\alpha} \uparrow$	$F_{\beta} \uparrow$	$E_{\xi} \uparrow$	$\mathcal{M} \downarrow$	$S_{\alpha} \uparrow$	$F_{\beta} \uparrow$	$E_{\xi} \uparrow$	$\mathcal{M} \downarrow$
$\Sigma = 1$	0.832	0.813	0.875	0.054	0.886	0.881	0.916	0.037
$\Sigma = 2$	0.836	0.816	0.873	0.052	0.888	0.889	0.922	0.035
$\Sigma = 3$	0.837	0.819	0.878	0.051	0.893	0.898	0.928	0.033

" $\Sigma = 1, 2, 3$ " mean BCD₁, BCD₁ + BCD₂ and BCD₁ + BCD₂ + BCD₃.

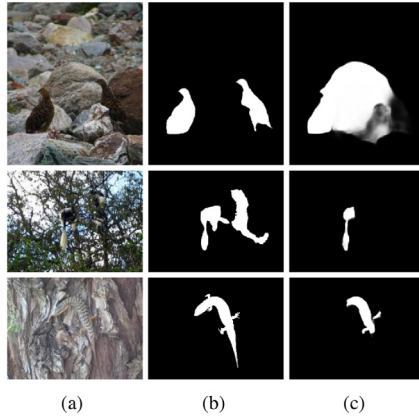


Fig. 10. Some failure cases of our method in extreme environments. (a) Image; (b) Ground truth; (c) Ours.

results of impact of BCD₁, BCD₂ and BCD₃ are listed in Table 5. We can observe that the most satisfactory results is get when deploying the scheme of " $\Sigma = 3$ ". Moreover, the enhancement by the " $\Sigma = 1$ ", " $\Sigma = 2$ " and " $\Sigma = 3$ " schemes is steadily incremental, indicating the significance of the trapezoidal network of BCD₁, BCD₂ and BCD₃.

4.4. Failure cases

Some failure cases of our method in extreme environments are shown in Fig. 10. In all these environments, the boundary contours seem to blend into the background noises. They look abnormally alike, which poses a disaster for the success of saliency detection. For example, the camouflage property of animals confuses the model into thinking that rocks are the salient object (i.e., 1st row), and it is hard to separate the salient objects from twigs and mountains (i.e., 2nd and 3rd rows). Solving these failure cases is a future direction of our work.

5. Conclusion

In this paper, we propose a novel ladder context correlation complementary network (LC³Net) for precise salient object detection. Different from the existing methods that ignore the gap between high level features and low level features, we aim to explicitly explore the reliable inherent complementarity of correlated contextual information. To assist the automatic collection of information on the diversity of initial features, we propose a filterable convolution block (FCB). To facilitate the intimate aggregation of different levels of features, we propose a dense cross module (DCM). To help the progressive shrinkage of multi-scale features from coarse to fine, we propose a bidirectional compression decoder (BCD). These three components of our method play a pivotal role in achieving remarkable detection results. Extensive experiments quantitatively and qualitatively demonstrate that the proposed LC³Net is superior to 20 state-of-the-arts.

CRedit authorship contribution statement

Xian Fang: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Visualization. **Jinchao Zhu:** Writing – review & editing, Visualization. **Xiuli Shao:** Writing – review & editing, Supervision. **Hongpeng Wang:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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